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Computer Vision – Project X proposal

As two study-abroad students living in Rome, we have had many opportunities to explore Rome's landmarks - the Colosseum, Trevi Fountain, Campo de' Fiori, and more. A recurring frustration, however, is familiar to any visitor. Nearly every photograph we take is crowded with strangers. Removing unwanted people or objects manually requires expensive tools, technical expertise, or reliance on paid cloud APIs. We believe this is a problem worth solving cheaply and automatically.

Object removal is a specific and practically important instance of image inpainting: given an image and a region to remove, the goal is to fill the masked area with a visually plausible and background-consistent reconstruction. Here are problems with existing solutions:
- They require large, models that do not run on consumer hardware or cheap GPUs.
- They require the user to manually draw a mask around the object which to remove.
- They require to manually write a text prompt.
- They are optimized for general inpainting, but we want object removal optimized.

We want to investigate whether a lightweight, fully automatic pipeline - requiring only a click on the object to remove - can produce competitive results compared to large, manually operated inpainting models.

Plan:
We plan to explore a pipeline that chains three components:
1. Automatic object segmentation, using a click-based segmentation model to generate the removal mask without manual drawing.
2. Automatic scene captioning using a vision-language model to generate a text description of the visible background, removing the need for a manual prompt.
3. Lightweight diffusion inpainting, adapting a diffusion-based inpainting approach, inspired by the RePaint resampling strategy, to a smaller and more accessible backbone than SD2, and studying whether targeted sampling refinements can compensate for the reduced model capacity.

The key research questions are: How much can we reduce model size before quality degrades unacceptably? Can automatic masking and prompting match manually provided inputs? What refinements matter most in the lightweight setting?

Dataset:
COCO 2017 val split -instance segmentation masks used as ground-truth removal targets for quantitative evaluation.
Our own photographs taken in Rome - used for qualitative demonstration and real-world validation.

Literature:
High-Resolution Image Synthesis with Latent Diffusion Models
[https://arxiv.org/abs/2112.10752]
RePaint: Inpainting using Denoising Diffusion Probabilistic Models
[https://arxiv.org/abs/2201.09865]